

Exercise on Prolog

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The following equations define the formal differentiation of a function:

$$\begin{aligned}\frac{dx}{dx} &= 1 & \frac{dy}{dx} &= 0 \quad \text{where } y \neq x \text{ or } y \in \mathbb{N} \\ \frac{d}{dx}(f + g) &= \frac{df}{dx} + \frac{dg}{dx} & \frac{d}{dx}(f - g) &= \frac{df}{dx} - \frac{dg}{dx} \\ \frac{d(-f)}{dx} &= -\frac{df}{dx} & \frac{d}{dx}(f \times g) &= f \times \frac{dg}{dx} + g \times \frac{df}{dx}\end{aligned}$$

Let us call **plus**, **minus** and **times** the relations for, respectively, addition (+), subtraction (−) and multiplication (×). Variables are defined by the predicate **var** and numbers by **num**. For instance, **minus(var(x),var(y))** denotes $x - y$; **minus(var(x))** denotes $-x$ (note that subtraction can have one or two arguments); **times(num(3),plus(var(x),var(x)))** means $3(x + x)$, etc.

Question. Define a relation **diff** with three arguments, the first being the function to be differentiated, the second the variable used for differentiation and the third is the derivative. For example **diff(F,var(x),DF)** means that DF is the derivative of F on variable x.

Hint. You will need the goal $\backslash+(X=Y)$ which means “X and Y do not match.”

Example. Now you can make queries like the following:

```
?- diff(plus(times(var(x),var(x)),var(y)), var(x), D).
D = plus(plus(times(var(x),num(1)),times(var(x),num(1))),num(0));
No
```

which proves that

$$\frac{d}{dx}(x \times x + y) = (x \times 1) + (x \times 1) + 0$$

Answer.

```
diff(var(X),var(X),num(1)). % 1
diff(num(_),var(_),num(0)). % 2
diff(var(Y),var(X),num(0)) :- \+(X=Y). % 3
diff(minus(F),X,minus(DF)) :- diff(F,X,DF). % 4
diff(plus(F,G),X,plus(DF,DG)) :- % 5
    diff(F,X,DF), diff(G,X,DG).
diff(minus(F,G),X,minus(DF,DG)) :- % 6
    diff(F,X,DF), diff(G,X,DG).
diff(times(F,G),X,plus(times(F,DG),times(G,DF))) :- % 7
    diff(F,X,DF), diff(G,X,DG).
```

Extra points. Give the trace of this example.

Answer.

